

Eenheid 3.3

Note Title

2016/04/06

Afgeleides van Trigonometrische
functies. (Stewart p 190)

Studiegids p 20

KEN BEWYSE!!

$$\bullet \frac{d}{dx} \sin x = \cos x$$

$$\bullet \frac{d}{dx} \cos x = -\sin x$$

$$\bullet \frac{d}{dx} \tan x = \sec^2 x$$

$$\bullet \frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

$$\bullet \frac{d}{dx} \sec x = \sec x \tan x$$

$$\bullet \frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

Stelling: Bewys dat $\frac{d}{dx} \cos x = -\sin x$

gebruik 2 hulpstellingen om dit te bewys:

Bewys:

• Hulpstelling A: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ ✓ "vaste" stelling (Eenheid 2.3)

? Hulpstelling B: ? $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$?

Bewys van B: $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$

Bewys: $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} \left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} \times \frac{\cos x + 1}{\cos x + 1}$$

$$= \lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x(\cos x + 1)}$$

Identiteit:
 $\sin^2 x + \cos^2 x = 1$
 $\cos^2 x - 1 = -\sin^2 x$

$$= \lim_{x \rightarrow 0} \frac{-\sin^2 x}{x(\cos x + 1)} \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} (-\sin^2 x) \times \frac{1}{x} \times \frac{1}{\cos x + 1}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \lim_{x \rightarrow 0} \sin x \times \lim_{x \rightarrow 0} \frac{1}{\cos x + 1}$$

$$= -1 \times 0 \times \frac{1}{2} = 0$$

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1}$$

Hulpstelling B

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$$

* Bewyse vir die formules van die afgeleides van trigonometriese funksies.

① $\frac{d}{dx}(\sin x) = \cos x$ (Stewart p191)

② $\frac{d}{dx}(\cos x)$ (Gebruik Hulpstelling 1 en 2)

$$= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$
$$= \lim_{h \rightarrow 0} \frac{\cos x \cosh - \sin x \sinh}{h} - \cos x$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{\cos x (\cosh - 1)}{h} - \lim_{h \rightarrow 0} \frac{\sin x \sinh}{h} \\
 &\stackrel{\text{L'Hôpital}}{=} \lim_{h \rightarrow 0} \cos x \cdot \lim_{h \rightarrow 0} \frac{(\cosh - 1)'}{h'} - \sin x \cdot \lim_{h \rightarrow 0} \frac{(\sinh)'}{h'} \\
 &= \cos x \times 0 - \sin x \times 1 = -\sin x
 \end{aligned}$$

③ $\frac{d}{dx} (\operatorname{cosec} x)$ (quotient rule)
Plan

$$= \frac{d}{dx} \left(\frac{1}{\sin x} \right)$$

$$= \frac{\frac{d}{dx}(1) \times \sin x - 1 \times \frac{d}{dx}(\sin x)}{(\sin x)^2}$$

$$= \frac{0 - \cos x}{(\sin x)^2} = - \left(\frac{\cos x}{\sin x} \right) \times \frac{1}{\sin x}$$

$$= -\operatorname{cosec} x (\cot x)$$

④ $\frac{d}{dx}(\cot x)$ (kwosientreeël)

$$= \frac{d}{dx} \left(\frac{\cos x}{\sin x} \right)$$

$$= \frac{\frac{d}{dx}(\cos x) \times \sin x - \cos x \times \frac{d}{dx}(\sin x)}{(\sin^2 x)}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = - \frac{(\sin^2 x + \cos^2 x)}{\sin^2 x}$$

$$= \frac{-1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x$$

Self studie:

$$\frac{d}{dx}(\sin x)$$

Hulpstelling A en B

$$\left(\begin{array}{c} \sin x \\ \frac{d}{dx}(\cos x) \end{array} \right)$$

$$\frac{d}{dx}(\sec x)$$

} kwasiëntieël

$$\frac{d}{dx}(\tan x)$$

Voorbeelde:

$$\textcircled{1} \frac{d}{dt}(\sqrt{t} + \sin t) = \frac{1}{2\sqrt{t}} + \cos t$$

of $= \frac{1}{2}t^{-\frac{1}{2}} + \cos t$

$$\textcircled{2} \frac{d}{dx} \left(\frac{4}{x^2} + x^3 \tan x \right)$$

..... - produktreël

$$\bullet = 4 \times (-2) x^{-3} + \left(3x^2 \tan x + x^3 \sec^2 x \right)$$

$$= -8x^{-3} + 3x^2 \tan x + x^3 \sec^2 x$$

$$\frac{d}{dx} \left(\frac{4}{x^2} \right) = 4 \times \frac{d}{dx} \left(x^{-2} \right)$$

$\overset{=k}{\textcircled{4}}$
 $\underline{\underline{=}}$

$$\textcircled{3} \frac{d}{du} \left(\frac{1 - 3 \sin u}{4 + 6 \cot u} \right)$$

kwosientreël

$$= \frac{(0 - 3 \cos u)(4 + 6 \cot u) - (1 - 3 \sin u)(0 - 6 \csc^2 u)}{(4 + 6 \cot u)^2}$$

x nul ✓

④ $\frac{d}{dz} (2 \cos z \sec z)$ Moenie vereenvoudig
nie!!

$= 2 \left[\frac{d}{dz} (\cos z \sec z) \right]$
produk reël

$= 2 \left[(-\sin z) \sec z + \cos z (\sec z \tan z) \right]$
✓ ✓